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Lab 07

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Age of Information and Reliability in IIoT Networks: A Practical Perspective

Age of Information (AoI) is more than a metric, but a lens. It reframes system responsiveness, shifting the focus from delivery to relevance. It's not about when data arrives, but how new it is at the moment it's used. That difference is so important in regards to time systems where a decision, even if backed by data, is only as good as the timeliness of that data! If a machine receives old information, the action it takes might already be outdated or even dangerous. This is especially true in Industrial Internet of Things (IIoT) environments where milliseconds can change outcomes.

Reading the study by Farag, Ali, and Stefanović (2023) helped me appreciate just how layered this topic really is. Their model is centered around a diversified IIoT network and shows the tug-of-war between maintaining updated information and dependable delivery. During the assignment I thought about how small technical adjustments like tweaking the transmission probability or observing packet loss can drastically shape the performance of an entire system. There's also the matter of different traffic types. Some demand immediacy, like in systems that regulate chemical mixtures. Others, like those depending on timed robotic movement, care more about responses within a precise window. Even timely data can become irrelevant if the action window has closed.

Working with the dataset of 10,000 IIoT communications I used Python to investigate how various features relate to AoI. Cleaning the data was fairly easy. I noticed quite a few infinite AoI entries and had to filter those out before moving on. From there, I dove into distributions, visual trends, and correlations to start forming hypotheses. It became immediately clear that transmission probability had a direct inverse relationship with AoI. Higher transmission rates meant fresher data, which made sense. Breaking it down further, traffic labeled as AoI sensitive had noticeably lower AoI scores than other types, reinforcing what I'd expected based on the theoretical classifications.

The heatmap gave me an early clue that packet loss probability (PLP) was a standout feature, and the Random Forest model later confirmed this. After preprocessing and scaling the features, I ran the model and evaluated it with metrics like MSE and R^2 , scoring an R^2 of 0.81 which felt solid given the complexity of the relationships involved. When I checked the feature importances, PLP dominated. Nearly 78% of the model's prediction accuracy could be traced back to it. While I had assumed things like the number of nodes might matter more, the model reminded me that theoretical relevance doesn't always translate into practical weight. To test what I was seeing, I simulated three network configurations with varying PLP and channel conditions. The results aligned exactly with the conceptual predictions: higher reliability led to lower AoI, and degraded conditions pushed AoI up.

What this exercise brought into focus for me is how related these variables really are and how tricky it can be to optimize them together. In the real world, engineers often have to pick their battles. Maximizing reliability sounds good in theory, but if the data arrives too late to matter, that reliability is flawed. I started thinking like a systems architect instead of just a data analyst. One approach that stood out was dynamic transmission scaling. These devices can increase their

broadcast frequency if packet loss worsens. Another promising angle is edge computing. By processing data locally and sending only the most urgent packets upstream, you can reduce congestion and keep AoI under control.

Two industries where these lessons could directly apply are smart grids and automated manufacturing. For smart grids, current measurements are essential to avoiding imbalances that could lead to outages. If data arrives stale, operators miss their window to act. Meanwhile, in robotic assembly lines, even the slightest delay in update communication can throw off synchronized movements and lead to production errors. Balancing freshness and reliability in these settings isn't optional!

What I appreciated most about this assignment was the tangible application that I could understand. Working through the data, building a model, and stress testing it against industry logic made the topic click in a different way. I wasn't just reading about trade-offs; I was experiencing them. And now, I'm thinking more critically about how data quality, timeliness, and reliability all shape the systems we rely on in everything from factories to energy grids. That's a perspective I want to carry forward in the remainder of this course.